



INTRODUCTION

An Introduction to Exercise and Sport Physiology

In this chapter and in the web study guide

Focus of Exercise and Sport Physiology

Acute and Chronic Responses to Exercise

The Evolution of Exercise Physiology

Beginnings of Anatomy and Physiology

Early History of Exercise Physiology

Era of Scientific Exchange and Interaction

Development of Contemporary Approaches

Integrative Physiology

Translational Physiology

Pioneering Women in Exercise Physiology

ACTIVITY 0.1 *Timeline* presents a historical perspective of the history of exercise physiology.

VIDEO 0.1 presents Jim Pawelczyk discussing the integration of cellular-level processes with a view of the entire organism.

Exercise Physiology in the 21st Century

Exercise in Personalized Medicine

The “-Omics” Revolution

Epigenetics

Bioinformatics

Exercise Physiology Beyond Earth’s Boundaries

VIDEO 0.2 presents Jim Pawelczyk discussing the four P’s of medicine and the important role of exercise in individualized health strategies.

Research: The Foundation for Understanding

The Research Process

Research Settings
Research Tools: Ergometers
Research Designs
Research Controls
Confounding Factors in Exercise Research
Units and Scientific Notation
Reading and Interpreting Tables and Graphs

ANIMATION FOR FIGURE 0.11 details the process of scientific research.

AUDIO FOR FIGURE 0.14 describes a cross-sectional study design.

AUDIO FOR FIGURE 0.15 describes a longitudinal study design.

ACTIVITY 0.2 *Interpreting Figures and Tables* explains the components of charts, figures, and tables and how to interpret their data.

AUDIO FOR FIGURE 0.16 describes how to interpret a line graph.

AUDIO FOR FIGURE 0.17 describes the nonlinear response pattern shown in the graph.

In Closing

Much of the history of exercise physiology in the United States can be traced to the effort of a Kansas farm boy, David Bruce (D.B.) Dill, whose interest in physiology first led him to study the composition of crocodile blood. Fortunately for what would eventually grow into the discipline of exercise physiology, this young scientist redirected his research to humans when he became the first research director of the Harvard Fatigue Laboratory in 1927. Throughout his life he was intrigued by the physiology and adaptability of many animals that survive extreme exercise and environmental conditions, but he is best remembered for his research on *human* responses to exercise, heat, high altitude, and other environmental factors. Dr. Dill always served as one of the human guinea pigs in his own studies. During the Harvard Fatigue Laboratory's 20-year existence, he and his coworkers produced approximately 350 scientific papers along with a classic book titled *Life, Heat, and Altitude*.¹⁰

After the Harvard Fatigue Laboratory closed its doors in 1947, Dr. Dill began a second career as deputy director of medical research for the Army Chemical Corps, a position he held until his retirement from that post in 1961. Dr. Dill was then 70 years old—an age he considered too young for retirement—so he moved his research to Indiana University, where he served as a senior physiologist until 1966. In 1967 he obtained funding to establish the Desert Research Laboratory at the University of Nevada at Las Vegas. Dr. Dill used this laboratory as a base for his studies on human tolerance to exercise in the desert and at high altitude. He continued his research and writing until his final retirement at age 93, the same year he produced his last publication, titled *The Hot Life of Man and Beast*.¹¹



Dr. David Bruce (D.B.) Dill (a) at the beginning of his career, (b) as director of the Harvard Fatigue Laboratory at age 42, and (c) at age 92 just before his fourth retirement.

The human body is an amazing machine. As you sit reading this introduction, countless perfectly coordinated and integrated events are occurring simultaneously in your body. These events allow complex functions, such as hearing, seeing, breathing, and information processing, to continue without any conscious effort. If you stand up, walk out the door, and jog around the block, almost all of your body's systems will be activated, enabling you to successfully shift from rest to exercise. If you continue this routine regularly for weeks or months and gradually increase the duration and intensity of your jogging, your body will adapt so that you can perform better. Therein lie the two basic components of the study of exercise physiology: the acute responses of the body to exercise in all its forms and the adaptation of the body's systems to repeated or chronic exercise, often called exercise training.

For example, as a point guard directs her team down the basketball court on a fast break, her body makes many adjustments that require a series of complex interactions involving many body systems. Adjustments occur even at the cellular and molecular levels. To enable the coordinated leg muscle actions as she moves rapidly down court, nerve cells from the brain, referred to as motor neurons, conduct electrical impulses down the spinal cord to the legs. On reaching the muscles, these neurons release chemical messengers that cross the gap between the nerve and muscle, each neuron exciting a number of individual muscle cells or fibers. Once the nerve impulses cross this gap, they spread along the length of each muscle fiber and attach to specialized receptors. Binding of the messenger to its receptor sets into motion a series of steps that activate the muscle fiber's contraction processes, which involve specific protein molecules—actin and myosin—and an elaborate energy system to provide the fuel necessary to sustain a single contraction and subsequent contractions. It is at this level that other molecules, such as adenosine triphosphate (ATP) and phosphocreatine (PCr), become critical for providing the energy necessary to fuel contraction.

In support of this sustained and rhythmic muscular contraction and relaxation, multiple additional systems are called into action, including the following:

- The skeletal system provides the basic framework around which muscles act.
- The cardiovascular system delivers fuel to working muscle and to all of the cells of the body and removes waste products.
- The cardiovascular and respiratory systems work together to provide oxygen to the cells and remove carbon dioxide.
- The integumentary system (skin) helps maintain body temperature by allowing the exchange of heat between the body and its surroundings.
- The nervous and endocrine systems coordinate this activity, while helping to maintain fluid and electrolyte balance and assisting in the regulation of blood pressure.

For centuries, scientists have studied how the human body functions at rest in health and disease. During the past 100 years or so, a specialized group of physiologists have focused their studies on how the body functions during physical activity and sport. This introduction presents a historical overview of exercise and sport physiology and then explains some basic concepts that form the foundation for the chapters that follow.

Focus of Exercise and Sport Physiology

Exercise and sport physiology have evolved from the fundamental disciplines of anatomy and physiology. Anatomy is the study of an organism's structure, or morphology. While anatomy focuses on the basic *structure* of various body parts and their interrelationships, **physiology** is the study of body *function*. Physiologists study how the body's organ systems, tissues, cells, and the molecules within cells work and how their functions are integrated to regulate the body's internal environment, a process called **homeostasis**. Because physiology focuses on the functions of body structures, understanding anatomy is essential to learning physiology. Furthermore, both anatomy and physiology rely on a working knowledge of biology, chemistry, physics, and other basic sciences.

Exercise physiology is the study of how the body's functions are altered when we are physically active, since exercise presents a

challenge to homeostasis. Because the environment in which one performs exercise has a large impact, **environmental physiology** has emerged as a subdiscipline of exercise physiology. **Sport physiology** further applies the concepts of exercise physiology to enhancing sport performance and optimally training athletes. Thus, sport physiology derives its principles from exercise physiology. Because exercise physiology and sport physiology are so closely related and integrated, it is often hard to clearly distinguish between them. Because the same underlying scientific principles apply, exercise and sport physiology are often considered together, as they are in this text.

Acute and Chronic Responses to Exercise

The study of exercise and sport physiology involves learning the concepts associated with two distinct exercise patterns. First, exercise physiologists are concerned with how the body responds to an individual bout of exercise, such as running on a treadmill for an hour or lifting weights. An individual bout of exercise is called **acute exercise**, and the responses to that exercise bout are referred to as acute responses. When examining the acute response to exercise, we are concerned with the body's immediate response to, and sometimes its recovery from, a single exercise bout.

The other major area of interest in exercise and sport physiology is how the body responds over time to the stress of repeated bouts of exercise, sometimes referred to as **chronic adaptation** or **training effects**. When one performs regular exercise over a period of days and weeks, the body adapts. The physiological adaptations that occur with chronic exposure to exercise or training improve both exercise capacity and efficiency. With resistance training, the muscles become stronger. With aerobic training, the heart and lungs become more efficient and endurance capacity of the muscles increases. As discussed later in this introductory chapter and in more detail in [chapters 10](#) and [11](#), these adaptations are highly specific to the type of training the person does.

In Review

- Exercise physiology evolved from its parent discipline, physiology. The two primary concerns of exercise physiology are
 - how the body responds to the acute stress of a single bout of exercise or physical activity; and
 - how the body adapts to the chronic stress of repeated bouts of exercise—that is, exercise training.
 - Some exercise physiologists use exercise or environmental conditions (heat, cold, altitude, and so on) to stress the body in ways that uncover basic physiological mechanisms. Others examine exercise training's effects on health, disease, and well-being. Sport physiologists apply these concepts to athletes and sport performance.
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The Evolution of Exercise Physiology

To students, contemporary exercise physiology may seem like a vast collection of new ideas never before subjected to rigorous scientific scrutiny. On the contrary, our current understanding of exercise physiology is based on the lifelong efforts of hundreds of outstanding scientists. The theories and hypotheses of modern physiologists have been shaped by the efforts of scientists who may be long forgotten. What we consider original or new is most often an assimilation of previous findings or the application of basic science to problems in exercise physiology. As with every discipline, there are, of course, a number of key scientists and pivotal scientific contributions that brought about significant advances in our knowledge of the physiological responses to exercise. The following section reflects on the history of the field of exercise physiology and on a few of the people who shaped it. It is impossible in this short section to do justice to the hundreds of pioneering scientists who paved the way and laid the foundation for modern exercise physiology.

Beginnings of Anatomy and Physiology

One of the earliest descriptions of human anatomy and physiology was Claudius Galen's Greek text *De fasciis*, published in the first century. As a physician to the gladiators, Galen had ample opportunity to study and experiment on human anatomy and was a great proponent of science based on observation and

experimentation. He was aware of the dire consequences of sedentary living and linked regular exercise to overall health and well-being by including regular exercise as one of his laws of health:

- Breathe fresh air.
- Eat the proper foods.
- Drink the right drinks.
- Exercise.
- Get adequate sleep.
- Have a daily bowel movement.
- Control your emotions.

Galen's theories of anatomy and physiology were so widely accepted that they remained unchallenged for nearly 1,400 years. Not until the 1500s were any truly significant contributions made to the understanding of both the structure and function of the human body. A landmark text by Andreas Vesalius, titled *Fabrica Humani Corporis* [*Structure of the Human Body*], presented his findings on human anatomy in 1543. Although Vesalius' book focused primarily on anatomical descriptions of various organs, he occasionally attempted to explain their functions as well. British historian Sir Michael Foster said, "This book is the beginning, not only of modern anatomy, but of modern physiology. It ended, for all time, the long reign of fourteen centuries of precedent and began in a true sense the renaissance of medicine" (p. 354).¹⁴

Most early attempts at explaining physiology were either incorrect or so vague that they could be considered no more than speculation. Attempts to explain how a muscle generates force, for example, were usually limited to a description of its change in size and shape during action because observations were limited to what could be seen with the naked eye. From such observations, Hieronymus Fabricius (ca. 1574) suggested that a muscle's contractile power resided in its fibrous tendons, not in its "flesh." Anatomists did not discover the existence of individual muscle fibers until Dutch scientist Antonie van Leeuwenhoek introduced the microscope (ca. 1660). How these fibers shorten and create force would remain a mystery

until the middle of the 20th century, when the intricate workings of muscle proteins could be studied by electron microscopy.

Early History of Exercise Physiology

Although exercise physiology is a relative newcomer to the world of science, one of its first publications appeared in 1793, when a paper by Séguin and Lavoisier described the oxygen consumption of a young man measured both in the resting state and while he repeatedly lifted a 7.3 kg (16 lb) weight for 15 min.²⁶ At rest the man used 24 L of oxygen per hour (L/h), which increased to 63 L/h during exercise. Lavoisier believed that the site of oxygen utilization and carbon dioxide production was in the lungs. Even though this concept was doubted by other physiologists of the time, it remained accepted doctrine until the middle of the 1800s, when several German physiologists demonstrated that combustion of oxygen occurred in cells throughout the entire body.

Although many advances in the understanding of circulation and respiration occurred during the 1800s, few efforts were made to focus on the physiology of physical activity. However, in 1888, an apparatus was described that enabled scientists to study subjects during mountain climbing, even though the subjects had to carry a 7 kg (15.4 lb) “gasometer” on their backs.³¹

Arguably the first published textbook on exercise physiology, *Physiology of Bodily Exercise*, was written in French by Fernand LaGrange in 1889.¹⁹ Considering the small amount of research on exercise that had been conducted up to that time, it is intriguing to read the author’s accounts of such topics as “Muscular Work,” “Fatigue,” “Habituation to Work,” and “The Office of the Brain in Exercise.” This early attempt to explain the response to exercise was, in many ways, limited to speculation and theory. Although some basic concepts of exercise biochemistry were emerging at that time, LaGrange was quick to admit that many details were still in the formative stages. For example, he stated that

Vital combustion [energy metabolism] has become very complicated of late; we may say that it is somewhat perplexed, and that it is difficult to give in a few words a clear and concise summary of it. It is a chapter of physiology which is being rewritten, and we cannot at this moment formulate our conclusions. (p. 395)¹⁹

Because the early text by LaGrange offered only limited physiological insights regarding bodily functions during physical activity, some argue that the third edition of a text by F.A. Bainbridge titled *The Physiology of Muscular Exercise*, published in 1931, should be considered the earliest scientific text on this subject.² Interestingly, that third edition was written by A.V. Bock and D.B. Dill, at the request of A.V. Hill, three key pioneers of exercise physiology discussed in this introductory chapter.

Archibald V. (A.V.) Hill was a significant figure in the history of exercise physiology. In his inaugural address as Jodrell Professor of Physiology at University College London, Hill stated the principles that subsequently shaped the field of exercise physiology:

It is strange how often a physiological truth discovered on an animal may be developed and amplified, and its bearings more truly found, by attempting to work it out on man. Man has proved, for example, far the best subject for experiments on respiration and on the carriage of gases by the blood, and an excellent subject for the study of kidney, muscular, cardiac and metabolic function.... Experiment on man is a special craft requiring a special understanding and skill, and “human physiology,” as it may be called, deserves an equal place in the list of those main roads which are leading to the physiology of the future. The methods, of course, are those of biochemistry, of biophysics, of experimental physiology; but there is a special kind of art and knowledge required of those who wish to make experiments on themselves and their friends, the kind of skill that the athlete and the mountaineer must possess in realizing the limits to which it is wise and expedient to go.

During the late 1800s, many theories were proposed to explain the source of energy for muscle contraction. Muscles were known to generate much heat during exercise, so some theories suggested that this heat was used directly or indirectly to cause muscle fibers to shorten. After the turn of the century, Walter Fletcher and Sir Frederick Gowland Hopkins observed a close relation between muscle action and lactate formation.¹² This observation led to the realization that energy for muscle action is derived from the breakdown of muscle glycogen to lactic acid (see [chapter 2](#)), although the details of this reaction remained obscure. Because of the high energy demands of exercising muscle, this tissue served as an ideal model to help unravel the mysteries of cellular metabolism. In 1921, A.V. Hill ([figure 0.1](#)) was awarded the Nobel Prize for his

findings on energy metabolism. At that time, biochemistry was in its infancy, although it was rapidly gaining recognition through the research efforts of such other Nobel laureates as Albert Szent-Györgyi, Otto Meyerhof, August Krogh, and Hans Krebs, all of whom were actively studying how living cells generate and use energy.

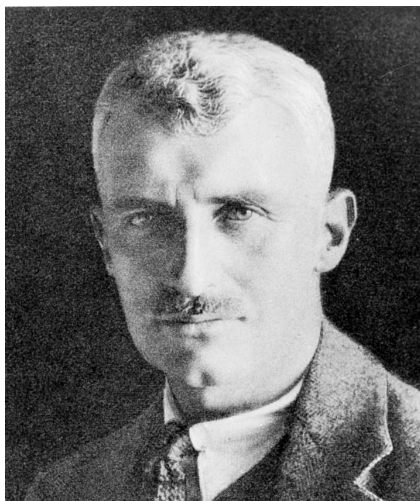


FIGURE 0.1 1921 Nobel Prize winner Archibald V. Hill (1927).

Although much of Hill's research was conducted with isolated frog muscle, he also conducted some of the first physiological studies of runners. Such studies were made possible by the technical contributions of John S. Haldane, who developed the methods and equipment needed to measure oxygen use during exercise. These and other investigators provided the basic framework for our understanding of whole-body energy production, which became the focus of considerable research during the middle of the 20th century and is incorporated into the manual and computer-based systems that are used to measure oxygen uptake in exercise physiology laboratories throughout the world today. In his address, A.V. Hill went on to acknowledge Haldane's contributions and discuss the wide range of applications he saw for his work in exercise physiology:

Quite apart from direct physiological research on man, the study of instruments and methods applicable to man, their standardization, their description, their reduction to routine, together with the setting up of standards of normality in man are bound to prove of great advantage to medicine; and not only to medicine but to all those activities and arts where normal man is the object of study. Athletics, physical training, flying, working,

submarines, or coal mines, all require a knowledge of the physiology of man, as does also the study of conditions in factories. The observation of sick men in hospitals is not the best training for the study of normal man at work. It is necessary to build up a sound body of trained scientific opinion versed in the study of normal man, for such trained opinion is likely to prove of the greatest service, not merely to medicine, but in our ordinary social and industrial life. Haldane's unsurpassed knowledge of the human physiology of respiration has often rendered immeasurable service to the nation in such activities as coal mining or diving; and what is true of the human physiology of respiration is likely also to be true of many other normal human functions.

Era of Scientific Exchange and Interaction

From the early 1900s through the 1930s, the medical and scientific environment in the United States was changing. This was an era of revolution in the education of medical students, led by changes at Johns Hopkins. More medical and graduate programs based their educational endeavors on the European model of experimentation and development of scientific insights. There were important advances in physiology in areas such as bioenergetics, gas exchange, and blood chemistry that served as the basis for advances in the physiology of exercise. Building on collaborations forged in the late 1800s, interactions of laboratories and scientists were promoted, and international meetings of organizations such as the International Union of Physiological Sciences created an atmosphere for free scientific exchange, discussion, and debate. Research laboratories and collaborations created during this period would go on to do some of the most important exercise physiology research of the 20th century.

Research on Athletes

For more than 100 years, athletes have served as subjects for study of the upper limits of human endurance. Perhaps the first physiological studies on athletes occurred in 1871. Austin Flint studied one of the most celebrated athletes of that era, Edward Payson Weston, an endurance runner-walker. Flint's investigation involved measuring Weston's energy balance (i.e., food intake versus energy expenditure) during Weston's attempt to walk 400 mi (644 km) in 5 days. Although the study resolved few questions about

muscle metabolism during exercise, it did demonstrate that some protein is lost from the body during prolonged heavy exercise.¹³

Throughout the 20th century, athletes were used repeatedly to assess the physiological capabilities of human strength and endurance and ascertain characteristics needed for record-setting performances. Some attempts have been made to use the technology and knowledge derived from exercise physiology to predict performance, prescribe training, or identify athletes with exceptional potential. In most cases, however, these applications of physiological testing are of little more than academic interest because few laboratory or field tests can accurately assess all the qualities required for someone to become a champion.

The Harvard Fatigue Laboratory

Perhaps no university has had more influence on the field of exercise physiology than Harvard. From 1891 to 1898, Harvard offered a degree in anatomy, physiology, and physical training under the direction of Dr. George Wells Fitz to “provide necessary knowledge about the science of exercise.” While that department changed its focus with Fitz’s departure in 1899, many other U.S. universities developed programs over the next 25 years that coupled basic science coursework with physical education.

A visit by A.V. Hill to Harvard University in 1926 had a significant impact on the founding and early activities of the Harvard Fatigue Laboratory (HFL), which was established a year later in 1927. Interestingly, the early home of the HFL was the basement of Harvard’s Business School, and its stated early mission was to conduct research on fatigue and other hazards in industry. Creation of this laboratory was due to the insightful planning of world-famous biochemist Lawrence J. (L.J.) Henderson. A young biochemist from Stanford University, David Bruce (D.B.) Dill, was appointed as the first director of research, a title Dill held until the HFL closed in 1947.

As noted earlier, Dill had aided Arlen “Arlie” Bock in writing the third edition of Bainbridge’s text on exercise physiology. Later in his career Dill credited the writing of that textbook with shaping the program of the HFL. Although he had little experience in applied human physiology, Dill’s creative thinking and ability to surround

himself with young, talented scientists created an environment that would lay the foundation for modern exercise and environmental physiology. For example, HFL personnel examined the physiology of endurance exercise and described the physical requirements for success in events such as distance running. Some of the most outstanding HFL investigations were conducted not in the laboratory but in the Nevada desert, on the Mississippi Delta, and in the White Mountains in California (with an altitude of 3,962 m, or 13,000 ft). These and other studies provided the foundation for future investigations on the effects of the environment on physical performance and in exercise and sport physiology.

In its early years, the HFL focused primarily on general problems of exercise, nutrition, and health. For example, the first studies on exercise and aging were conducted in 1939 by Sid Robinson (see [figure 0.2](#)), a student at the HFL. On the basis of his studies of subjects ranging in age from 6 to 91 years, Robinson described the effect of aging on maximal heart rate and oxygen uptake.²⁴ But with the onset of World War II, Henderson and Dill realized the HFL's potential contribution to the war effort, and research at the HFL took a different direction. Harvard Fatigue Lab scientists and support personnel were instrumental in forming new laboratories for the Army, Navy, and Army Air Corps (now the Air Force). They also published the methodologies necessary for relevant military research, methods that are still in use throughout the world.

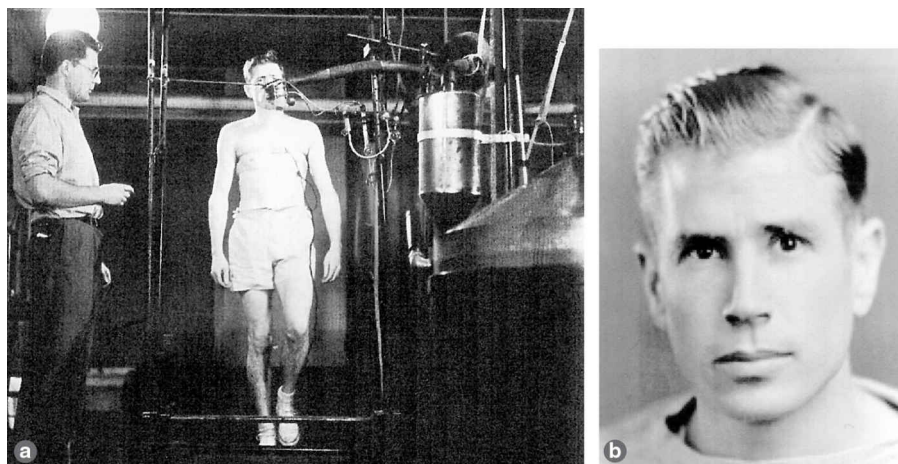


FIGURE 0.2 Sid Robinson (a) being tested by R.E. Johnson on the treadmill in the Harvard Fatigue Laboratory and (b) as a Harvard student and athlete in 1938.

Today's exercise physiology students would be amazed at the methods and devices used in the early days of the HFL and at the time it took to conduct research projects in those days. What is now accomplished in milliseconds with the aid of computers and automatic analyzers literally demanded days of effort by HFL scientists. Measurements of oxygen uptake during exercise, for example, required collecting expired air in Douglas bags and analyzing it for oxygen and carbon dioxide by using a manually operated chemical analyzer, without the help of a computer, of course (see [figure 0.3](#)). The analysis of a single 1 min sample of expired air required 20 to 30 min of effort by one or more laboratory workers. Today, scientists make such measurements almost instantaneously and with little physical effort. One must marvel at the dedication, diligence, and hard work of the HFL's exercise physiology pioneers. Using the equipment and methods available at the time, HFL scientists published approximately 350 research papers over a 20-year period.

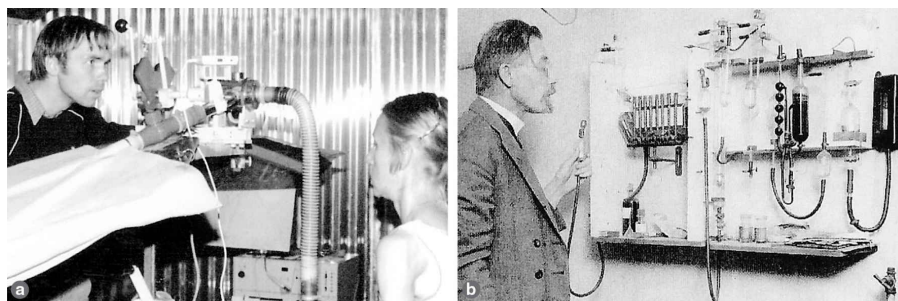


FIGURE 0.3 (a) Early measurements of metabolic responses to exercise required the collection of expired air in a sealed bag known as a Douglas bag. (b) A sample of that gas then was measured for oxygen and carbon dioxide using a chemical gas analyzer, as illustrated by this photo of Nobel laureate August Krogh.

The HFL was an intellectual environment that attracted young physiologists and physiology doctoral students from all over the globe. Scholars from 15 countries worked in the HFL between 1927 and its closure in 1947. Most went on to develop their own laboratories and become noteworthy figures in exercise physiology in the United States, including Sid Robinson, Henry Longstreet Taylor, Lawrence Morehouse, Robert E. Johnson, Ancel Keys, Steven Horvath, C. Frank Consolazio, and William H. Forbes.

Notable international scientists who spent time at the HFL included August Krogh, Lucien Brouha, Edward Adolph, Walter B. Cannon, Peter Scholander, and Rodolfo Margaria, along with several other notable Scandinavian scientists discussed later. Thus, the HFL planted seeds of intellect at home and around the world that resulted in an explosion of knowledge and interest in this new field. Most contemporary exercise physiologists can trace the roots of their research training back to the HFL.

Scandinavian Influence

In 1909, Johannes Lindberg established a laboratory that became a fertile breeding ground for scientific contributions at the University of Copenhagen in Denmark. Lindberg and 1920 Nobel Prize winner August Krogh teamed up to conduct many classic experiments and published seminal papers on topics ranging from the metabolic fuels for muscle to gas exchange in the lungs. This work was continued from the 1930s into the 1970s by Erik Hohwü-Christensen, Erling Asmussen, and Marius Nielsen.

As a result of contacts between D.B. Dill and August Krogh, these three Danish physiologists came to the HFL in the 1930s, where they studied exercise in hot environments and at high altitude. After returning to Europe, each man established a separate line of research. Asmussen and Nielsen became professors at the University of Copenhagen, where Asmussen studied the mechanical properties of muscle and Nielsen conducted studies on control of body temperature. Both remained active at the University of Copenhagen's August Krogh Institute until their retirements.

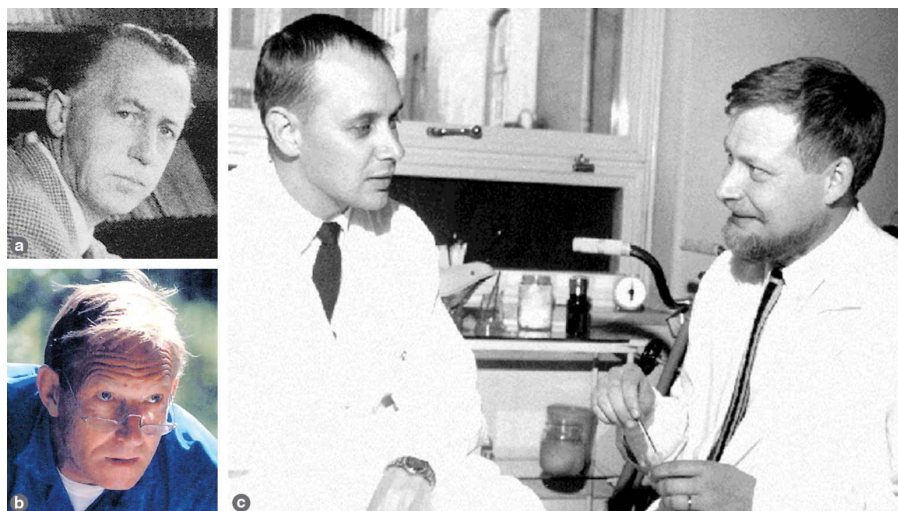


FIGURE 0.4 (a) Erik Hohwü-Christensen was the first physiology professor at the College of Physical Education at Gymnastik-och Idrottshögskolan in Stockholm, Sweden. (b) Bengt Saltin, winner of the 2002 Olympic Prize. (c) Jonas Bergstrom (left) and Eric Hultman (right) were the first to use muscle biopsy to study muscle glycogen use and restoration before, during, and after exercise.

In 1941, Hohwü-Christensen (see [figure 0.4a](#)) moved to Stockholm, Sweden, to become the first physiology professor at the College of Physical Education at Gymnastik-och Idrottshögskolan (GIH). In the late 1930s, he teamed with Ole Hansen to conduct and publish a series of five studies of carbohydrate and fat metabolism during exercise. These studies are still cited frequently and are considered to be among the first and most important sport nutrition studies. Hohwü-Christensen introduced Per-Olof Åstrand to the field of exercise physiology. Åstrand, who conducted numerous studies related to physical fitness and endurance capacity during the 1950s and 1960s, became the director of GIH after Hohwü-Christensen retired in 1960. During his tenure at GIH, Hohwü-Christensen mentored a number of outstanding scientists, including Bengt Saltin, who was the 2002 Olympic Prize winner for his many contributions to the field of exercise and clinical physiology (see [figure 0.4b](#)).

In addition to their work at GIH, both Hohwü-Christensen and Åstrand interacted with physiologists at the Karolinska Institute in Stockholm, Sweden, who studied clinical applications of exercise. It is hard to single out the most exceptional contributions from this institute, but Jonas Bergstrom's ([figure 0.4c](#)) reintroduction of the biopsy needle (ca. 1966) to sample muscle tissue was a pivotal point

in the study of human muscle biochemistry and muscle nutrition. This technique, which involves withdrawing a tiny sample of muscle tissue with a needle inserted into the muscle through a small incision, was originally introduced in the early 1900s to study muscular dystrophy. The needle biopsy enabled physiologists to conduct histological and biochemical studies of human muscle before, during, and after exercise.

Other invasive studies of blood circulation were subsequently conducted by physiologists at GIH and at the Karolinska Institute. Just as the HFL had been the mecca of exercise physiology research between 1927 and 1947, the Scandinavian laboratories were equally noteworthy beginning in the late 1940s. Many leading investigations over the following 35 years were collaborations between American and Scandinavian exercise physiologists. Norwegian Per Scholander introduced a gas analyzer in 1947. Finn Martti Karvonen published a formula for calculating exercise heart rate that is still widely used today.

Other Research Milestones

Physiology has always been the basis for clinical medicine. In the same way, exercise physiology has provided essential knowledge for many other areas, such as physical education, physical fitness, physical therapy, and health promotion. In the late 1800s and early 1900s, physicians such as Amherst College's Edward Hitchcock, Jr. and Harvard's Dudley Sargent studied body proportions (anthropometry) and the effects of physical training on strength and endurance. Although a number of physical educators introduced biological science to the undergraduate physical education curriculum, Peter Karpovich, a Russian immigrant who had been briefly associated with the HFL ([figure 0.5a](#)), played a major role in introducing physiology to physical education. Karpovich established his own research facility and taught physiology at Springfield College (Massachusetts) from 1927 until his death in 1968.

Although he made numerous contributions to physical education and exercise physiology research, he is best remembered for the outstanding students he advised, including Charles Tipton and

Loring Rowell, both recipients of the American College of Sports Medicine Honor and Citation Awards.

Another Springfield faculty member, swim coach Thomas K. (T.K.) Cureton ([figure 0.5b](#)), created an exercise physiology laboratory at the University of Illinois at Urbana-Champaign in 1941. He continued his research and taught many of today's leaders in physical fitness and exercise physiology until his retirement in 1971. Physical fitness programs developed by Cureton and his students, as well as Kenneth Cooper's 1968 book, *Aerobics*, established a physiological rationale for using exercise to promote a healthy lifestyle.⁹

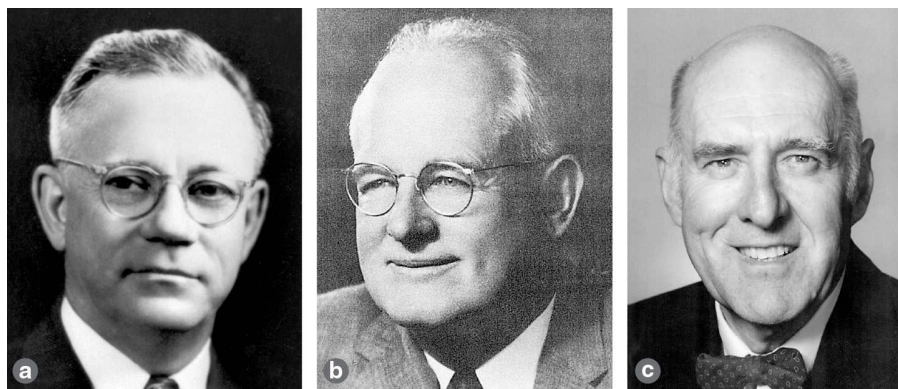


FIGURE 0.5 (a) Peter Karpovich introduced the field of exercise physiology during his tenure at Springfield College. (b) Thomas K. Cureton directed the exercise physiology laboratory at the University of Illinois at Urbana-Champaign from 1941 to 1971. (c) At Penn State, Elsworth Buskirk founded an intercollege graduate program focusing on applied physiology (1966) and constructed The Laboratory for Human Performance Research (1974).

Another contributor to the establishment of exercise physiology as an academic endeavor was Elsworth R. “Buz” Buskirk ([figure 0.5c](#)). After holding positions as chief of the environmental physiology section at the Quartermaster Research and Development Center in Natick, Massachusetts (1954-1957), and research physiologist at the National Institutes of Health (1957-1963), Buskirk moved to Pennsylvania State University, where he stayed for the remainder of his career. At Penn State, Buz founded the Intercollege Graduate Program in Physiology (1966) and constructed The Laboratory for Human Performance Research (1974), the nation's first freestanding research institute devoted to the study of human adaptation to

exercise and environmental stress. He remained an active scholar until his death in April of 2010.

Although there was some awareness as early as the mid-1800s of a need for regular physical activity to maintain optimal health, this idea did not gain popular acceptance until the late 1960s. Subsequent research has continued to support the importance of exercise in slowing the physical decline associated with aging, preventing or mitigating the problems associated with chronic diseases, and rehabilitating injuries.

Development of Contemporary Approaches

Much advancement in exercise physiology must be credited to improvements in technology. In the late 1950s, Henry L. Taylor and Elsworth R. Buskirk published two seminal papers^{6,28} describing the criteria for measuring maximal oxygen uptake and establishing that measure as the gold standard for cardiorespiratory fitness. In the 1960s, development of electronic analyzers to measure respiratory gases made studying energy metabolism much easier and more productive than before. This technology and radio telemetry (which uses radio-transmitted signals), used to monitor heart rate and body temperature during exercise, were developed as a result of the U.S. space program. Although such instruments took much of the labor out of research, they did not alter the direction of scientific inquiry. Until the late 1960s, most exercise physiology studies focused on the whole body's response to exercise. The majority of investigations involved measurements of such variables as oxygen uptake, heart rate, body temperature, and sweat rate. Cellular responses to exercise received little attention.

Biochemical Approaches

In the mid-1960s, three biochemists emerged who were to have a major impact on the field of exercise physiology. John Holloszy ([figure 0.6a](#)) at Washington University in St. Louis, Missouri, Charles "Tip" Tipton ([figure 0.6b](#)) at the University of Iowa, and Phil Gollnick ([figure 0.6c](#)) at Washington State University first used rats and mice to study muscle metabolism and examine factors related to fatigue. Their publications and their training of graduate and postdoctoral students resulted in a more biochemical approach to exercise

physiology research. Holloszy was ultimately awarded the 2000 Olympic Prize for his contributions to exercise physiology and health.

Before the 1960s, there were few biochemical studies on the adaptations of muscle to training. Although the field of biochemistry can be traced to the early part of the 20th century, this special area of chemistry was not applied to human muscle until Bergstrom and Hultman reintroduced and popularized the needle biopsy procedure in 1966. Initially, this procedure was used to examine glycogen depletion during exhaustive exercise and its resynthesis during recovery. In the early 1970s, as noted earlier, a number of exercise physiologists used the muscle biopsy method, histological staining, and the light microscope to determine human muscle fiber types.

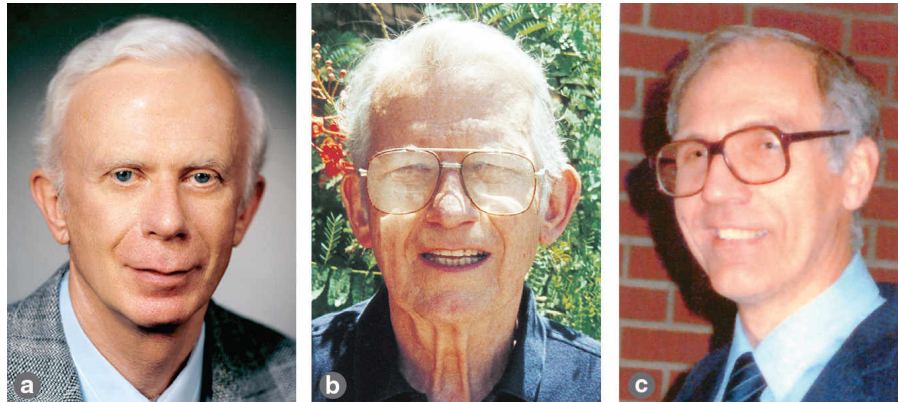


FIGURE 0.6 (a) John Holloszy was the winner of the 2000 Olympic Prize for scientific contributions in the field of exercise science. (b) Charles Tipton was a professor at the University of Iowa and the University of Arizona and a mentor to many students who have become the leaders in molecular biology and genomics. (c) Phil Gollnick conducted muscle and biochemical research at Washington State University.

Around the time Bergstrom reintroduced the needle biopsy procedure, exercise physiologists who were well trained as biochemists emerged. In Stockholm, Bengt Saltin realized the value of this procedure for studying human muscle structure and biochemistry. He first collaborated with Bergstrom in the late 1960s to study the effects of diet on muscle endurance and muscle nutrition. About the same time, Reggie Edgerton (University of California at Los Angeles) and Phil Gollnick were using rats to study the characteristics of individual muscle fibers and their responses to training. Saltin subsequently combined his knowledge of the biopsy

procedure with Gollnick's biochemical talents. These researchers were responsible for many early studies on human muscle fiber's characteristics and use during exercise. Although many biochemists have used exercise to study metabolism, few have had more impact on the current direction of human exercise physiology than Bergstrom, Saltin, Tipton, Holloszy, and Gollnick.

Other Tools and Techniques

The history of exercise physiology has, in some ways, been driven by advancements in technologies adapted from basic sciences. The early studies of energy metabolism during exercise were made possible by the invention of gas-collecting equipment and chemical analysis of oxygen and carbon dioxide. Chemical determination of blood lactic acid seemed to provide some insights regarding the aerobic and anaerobic aspects of muscular activity, but these data told us little regarding the production and removal of this by-product of exercise. Likewise, blood glucose measurements taken before, during, and after exhaustive exercise proved to be interesting data but were of limited value for understanding the energy exchange at the cellular level.

Over the last 30 years, muscle physiologists have used various chemical procedures to understand how muscles generate energy and adapt to training. Test tube experiments (in vitro) with muscle biopsy samples have been used to measure muscle proteins (enzymes) and determine the muscle fiber's capacity to use oxygen. Although these studies provided a snapshot of the fiber's potential to generate energy, they often left more questions than answers. It was natural, therefore, for the sciences of cell biology to move to an even deeper level. It was apparent that the answers to those questions must lie within the fiber's molecular makeup.

Although not a new science, molecular biology has become a useful tool for exercise physiologists who wish to delve more deeply into the cellular regulation of metabolism and adaptations to the stress of exercise. Physiologists like Frank Booth and Ken Baldwin ([figure 0.7](#)) have dedicated their careers to understanding the molecular regulation of muscle fiber characteristics and function and have laid the groundwork for our current understanding of the

genetic controls of muscle growth and atrophy. The use of molecular biological techniques to study the contractile characteristics of single muscle fibers is discussed in [chapter 1](#).

Well before James Watson and Francis Crick unraveled the structure of deoxyribonucleic acid (DNA) in 1953, scientists appreciated the importance of genetics in predetermining the structure and function of all living organisms. The newest frontier in exercise physiology combines the study of molecular biology and genetics. Since the early 1990s, scientists have attempted to explain how exercise causes signals that affect the expression of genes within skeletal muscle.

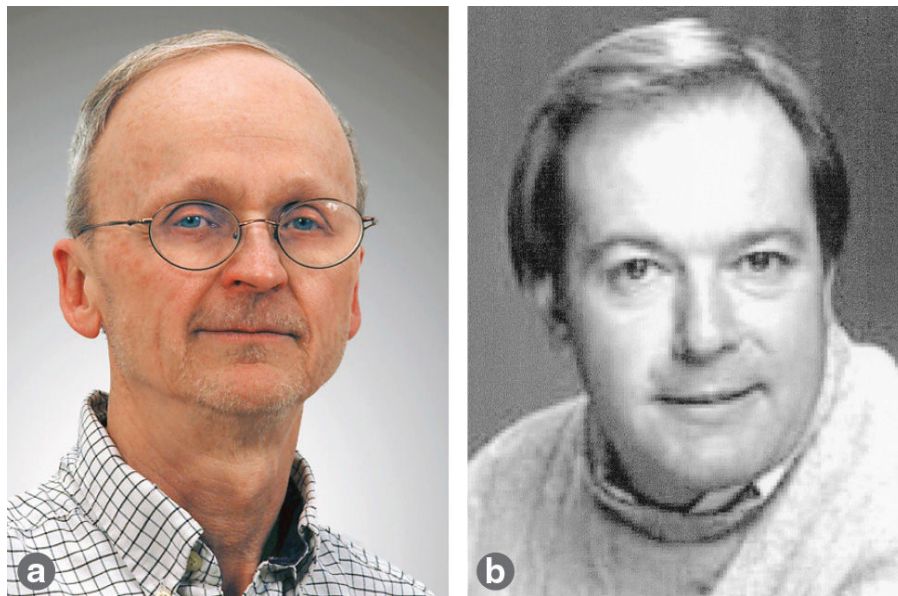


FIGURE 0.7 (a) Frank Booth and (b) Ken Baldwin.

In retrospect, it is apparent that since the beginning of the 20th century, the field of exercise physiology has evolved from measuring whole-body function (i.e., oxygen consumption, respiration, and heart rate) to molecular studies of muscle fiber genetic expression. There is little doubt that exercise physiologists of the future will need to be well grounded in biochemistry, molecular biology, and genetics.

Integrative Physiology

VIDEO 0.1 Presents Jim Pawelczyk discussing the integration of cellular-level processes with a view of the entire organism.



With the announcement of the sequencing of the human genome in 2001, it was hoped that one day, scientists could simply analyze cheek cells from a mouth swab and, by looking at your gene sequence, predict whether you were at risk for developing diabetes or cardiovascular disease.^{7,8} More promising was the idea that detection of these predictive genetic variations could aid in developing more effective treatments for these debilitating diseases.

These advances in biotechnology have produced huge volumes of data over the past several years, but the initial optimism regarding the prediction and treatment of human disease has not been fulfilled.¹⁷ While there are a few specific gene mutations that have reliable predictive power, such as the breast cancer gene BRCA1, the translation of genetic technologies into predictive diagnostics or therapies has largely not occurred. In fact, analysis of traditional risk factors still has much more predictive power for evaluating risk for type 2 diabetes than the evaluation of genetic risk scores based on 20 different gene variants associated with this disease.²⁷

In the era of mega-genomic data, where does the study of physiology fit in? And is it still relevant to human health and disease? One outspoken advocate for the field of **integrative physiology** is Dr. Michael J. Joyner. Dr. Joyner is an award-winning, distinguished investigator at the Mayo Clinic who has critically questioned the functional value of so-called reductionist thinking in molecular biology. In contrast to examining biological processes at the lowest

common level (for example, how genes code for proteins in cells), integrative physiology examines how whole organisms function and adapt to internal and external stresses (including exercise). This approach is informed by the concepts of homeostasis, regulated organ systems, and redundancy in physiological systems. Moreover, integrative physiologists strive to ask hypothesis-driven research questions and design defensible experiments to test those hypotheses.

The importance of seeking to study biological questions from an integrative, regulated approach is highlighted by the influences of culture, environment, and behavior on disease pathology. The challenge for integrative physiologists is to incorporate key findings from genetics and molecular biology and to examine how behavioral patterns, including physical activity, diet, and stress, interplay with this genetic variation to affect health and disease.

Translational Physiology

Exercise physiologists, by the nature of the topics we study and the variety of approaches we use in those studies, make valuable contributions to what has become known as **translational physiology**. *Translational physiology* is a term that was originally used in the early 1990s to refer to the research process needed to link cancer risk with its predisposing genetic factors.²⁵ The field of translational physiology has broadened substantially from that time to include the processes by which basic research findings are extended to the clinical research setting, then to the realm of clinical practice, and finally to health policy ([figure 0.8](#)). This translational research continuum, however, works best in a bidirectional manner, such that population-based problems, like obesity, also drive the basic research questions that exercise physiologists ask. In turn, these basic research findings eventually drive changes in clinical practice and overall community health.

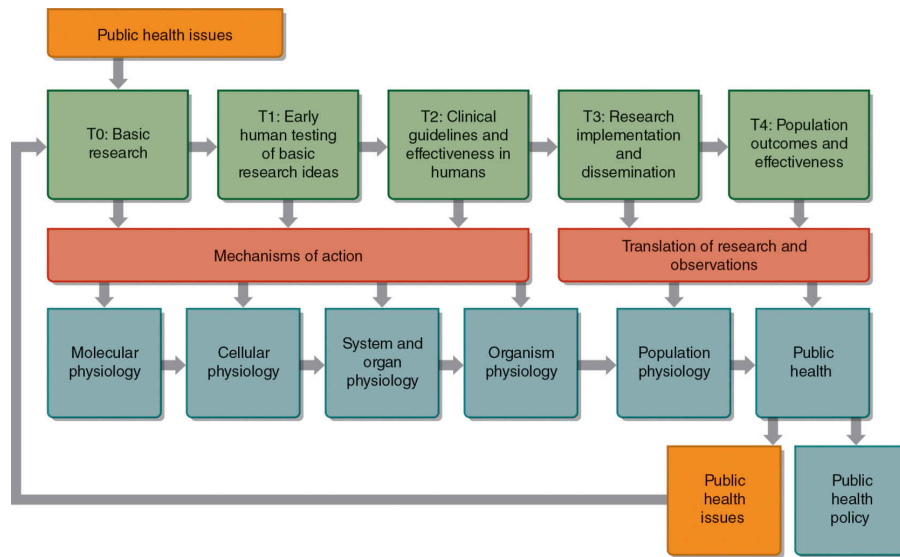


FIGURE 0.8 Flow chart for translational physiology.

Adapted from Seals (2013).

A good example of opportunities in translational physiology is in the field of aging. Advancing age by itself is a risk factor for many chronic diseases and presents a significant challenge to our health care system and to society in general. In order to fully understand the underlying physiology of aging and be able to engage in appropriate interventions to keep the aging population healthy, we must understand the aging process from the molecular level all the way to the community and population levels. Being able to successfully contribute to the translational physiology process requires a broad skill set to critically examine data and approach scientific problems with new goals in mind, from bench to bedside and from bedside to community.

In Review

- In an era that seems to stress a reductionist (genes, molecules) approach to science, there is an acute need for exercise physiologists to continue to study biological questions from an integrative, hypothesis-driven approach.
- The field of translational physiology addresses the processes by which basic research findings are extended to the clinical research setting, then to the realm of clinical practice, and finally to health policy.

Pioneering Women in Exercise Physiology

While outstanding female exercise physiologists are now commonplace, as in many areas of science, the contributions of women to exercise physiology were slow to gain recognition. In 1954, Irma Rhyming collaborated with her future husband, P.-O. Åstrand, to publish a classic study that provided a means to predict aerobic capacity from submaximal heart rate.¹ Although this indirect method of assessing physical fitness has been challenged over the years, its basic concept is still in use today.

In the 1970s, two Swedish women, Birgitta Essén and Karen Piehl (figure 0.9), gained international attention for their research on human muscle fiber composition and function. Essen, who collaborated with Bengt Saltin, was instrumental in adapting microbiological methods to study the small amounts of tissue obtained with the needle biopsy procedure. Her efforts enabled others to conduct studies on the muscle's use of carbohydrates and fats and to identify different muscle fiber types. Piehl published a number of studies that illustrated which muscle fiber types were activated during both aerobic and anaerobic exercise.

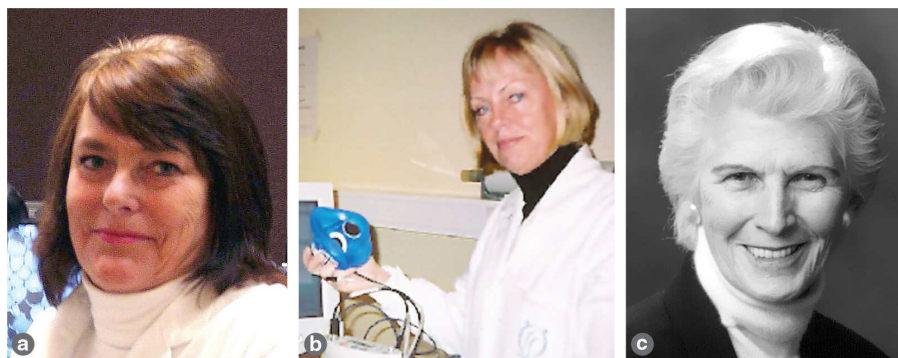


FIGURE 0.9 (a) Birgitta Essén collaborated with Bengt Saltin and Phil Gollnick in publishing the earliest studies on muscle fiber types in human muscle. (b) Karen Piehl was among the first physiologists to demonstrate that the nervous system selectively recruits type I (slow-twitch) and type II (fast-twitch) fibers during exercise of differing intensities. (c) Barbara Drinkwater was among the first to conduct studies on female athletes and to address issues specifically related to the female athlete.

In the 1970s and 1980s, a third Scandinavian female physiologist, Bodil Nielsen, daughter of Marius Nielsen, actively conducted studies on human responses to environmental heat stress and dehydration. Her studies even encompassed measurements of body temperature during immersion in water. At about the same time, an

American exercise physiologist, Barbara Drinkwater ([figure 0.9c](#)), was doing similar work at the University of California at Santa Barbara. Her studies were often conducted in collaboration with Steven Horvath, D.B. Dill's son-in-law and director of the UCSB's environmental physiology laboratory. Drinkwater's contributions to environmental physiology and study of the physiological problems confronting the female athlete gained international recognition. In addition to their scientific contributions, the legacy of these and other women in physiology includes the credibility they earned and the roles they played in attracting other young women to the fields of exercise physiology and medicine.

The intent of this section has been to provide readers with an overview of the personalities and technologies that have helped to shape the field of exercise physiology. Naturally, a comprehensive review of all the scientists and research associated with this field is not possible in a text intended as an introduction to exercise physiology, but for those students who wish to take an in-depth look at the historical background in exercise physiology, there are several good sources. Now that we understand the historical basis for the discipline of exercise physiology, from which sport physiology emerged, we can explore some basic principles of, and tools used in, exercise and sport physiology.

Exercise Physiology in the 21st Century

The field of exercise physiology is rapidly evolving. Ever-expanding technological developments and new approaches to science have substantial implications for health, medicine, and biomedical research. Exercise physiology and our understanding of the physiological processes that underpin physical activity are often at the forefront of this new age of science.

Exercise in Personalized Medicine

VIDEO 0.2 Presents Jim Pawelczyk discussing the four P's of medicine and the important role of exercise in individualized health strategies.



In 2007, the United States Congress passed the Genomics and Personalized Medicine Act. The intent of this legislation was to implement and support research related to formulating a personalized prescription to fit each patient's unique genetic and environmental characteristics in order to optimize health care strategies.^{15,16} This personalized medicine concept first emerged in the field known as pharmacogenomics, which provided scientific insights into why some individuals respond favorably to certain drugs while others do not respond (or may even respond adversely). For example, studies have identified two different genes that influence an individual's ability to metabolize the blood thinner warfarin and make it possible for a physician to prescribe an appropriate dosage to optimize the drug's therapeutic effectiveness for each individual patient.²⁹

Similarly, there has been a recent push to personalize each individual's exercise prescription.⁵ Exercise is a powerful intervention for the treatment of many different medical conditions—cardiovascular disease, diabetes mellitus, osteoporosis, metabolic diseases, and many more. However, there is significant heterogeneity or variability in people's abilities to perform exercise and adapt to the effects of exercise training,²¹ especially in individuals with different clinical disease manifestations. Moreover, researchers are just beginning to understand and formulate optimal training programs or personalized dosages of exercise to produce beneficial responses in these patients.

Researchers are designing experimental paradigms to determine (1) the mechanisms through which exercise produces effects (either positive or negative) on a cellular and systems level, (2) the optimal

dosage of exercise to produce results in different clinical populations, (3) the best way to evaluate a person's responses to exercise on an individual and a group level, and (4) the benefit of adding exercise therapy to existing disease treatment strategies. Part of the challenge in personalizing exercise medicine is to understand, on a genomic and systems level, the mechanisms responsible for the huge variability in individuals' responses to exercise training. Eventually, the long-term outcomes from large randomized clinical trials examining intraindividual variability in responses to exercise in humans will make it possible to develop *personalized* strategies to be implemented in preventive health care interventions,⁵ including the health benefits of regular exercise.

The “-Omics” Revolution

As a part of the Human Genome Project, scientists sequenced all 3.2 billion nucleotides that compose the human genome. This was mostly completed in 2003 (the last chromosome was sequenced in 2006) at an estimated cost of \$2.7 billion. Today, the entire human genome can be sequenced for less than \$1,000. This has opened up new fields of science, often called *-omics*. This new research area in turn has fostered the development of new technologies aimed at the universal detection of gene sequences and variants (genomics), the expression of genes at the messenger RNA (mRNA) level (transcriptomics), the proteins that are produced (proteomics), and other products of metabolic reactions (i.e., metabolites, studied in metabolomics)³⁰ involved in all aspects of physiological function (see [figure 0.10](#)). One purported appeal of *-omics* research is that a highly complex system (e.g., the exercising human) can be more fully understood if it is examined at each of the most basic levels of inquiry. As it is applied to exercise physiology, the primary goal of *-omics* research is to illuminate exercise physiology and behavior in order to better understand the preventive and therapeutic values of exercise.⁴

Exercise **genomics** research examines the role of individual (or groups of) genes in modifying the impact of exercise training and physical activity on performance and health- and fitness-related traits. This is based on accumulating evidence that variations within

the DNA sequences (called *single nucleotide polymorphisms*, or SNPs) of one or more genes may contribute to differences in exercise behavior, cardiorespiratory and muscular fitness, cardiovascular and metabolic function during acute exercise, and adaptations to exercise training.³

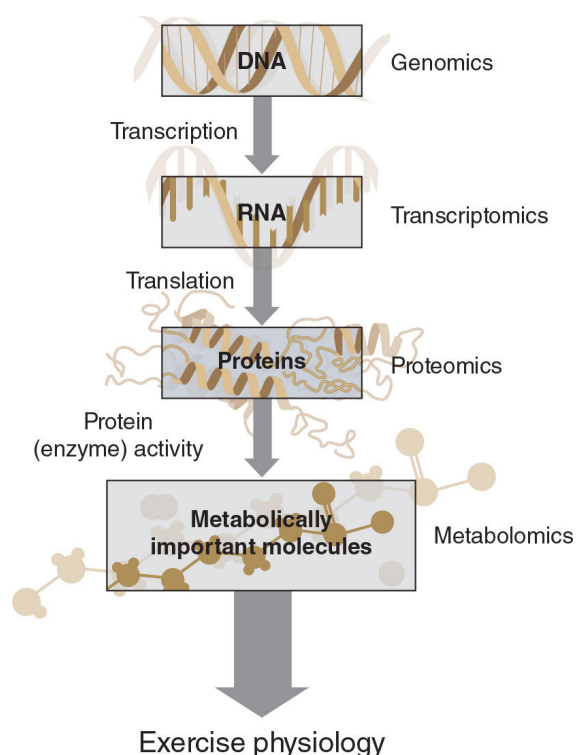


FIGURE 0.10 The link between genomics, transcriptomics, proteomics, and metabolomics in the context of exercise physiology.

Using genomics approaches, researchers are also attempting to examine the genetic basis of these highly complex traits by examining tissue-specific mRNA levels. The technologies used to confirm a gene target and define its biological function are increasingly sophisticated and now include DNA and RNA sequencing, *in vitro* cell-based investigations, genetic modifications in animal models, and selective breeding of animals for extreme performance traits to identify target genes and their variants.

More recently, researchers have started to combine genomics and transcriptomics. That is, by examining the RNA strands produced during transcription (transcript abundance) in relevant tissues, researchers can predict a certain trait and identify gene targets for

subsequent genomics research. These new gene targets can then be probed for their DNA sequence variants and their relation with other traits of interest. This integrated strategy within the -omics world has the potential to expand our understanding of exercise physiology at a level of detail that was not possible in the past. For example, understanding the exercise training–induced alterations in gene expression may provide novel candidates for genomics and genetics research aimed at further understanding the physiology of exercise.³

Exercise proteomics aims to study the entire protein content of a biological tissue in a particular situation (e.g., immediately after a resistance training session) or over a predetermined period (e.g., before and after months of endurance training), enabling investigators to examine the molecular mechanisms that underlie physiological adaptations to exercise.²² The original tool for proteomic analysis was a procedure called two-dimensional polyacrylamide gel electrophoresis. During the current genomic research era, these gel-based methodologies have continued to improve and are now being coupled with newer techniques based on protein labeling, peptide fragmentation, and high-throughput mass spectrometry to improve proteomic analyses. Research combining proteomic data with the genomic approaches described previously will continue to expand our understanding of exercise physiology by providing a big picture of how exercise affects the various organs and systems in the body to improve physiological function, exercise performance, and health in general.

Epigenetics

It is now apparent that exercise alters gene expression, expression of transcription factors, and other regulatory proteins. These exercise-induced alterations have functional consequences at many levels, including metabolism, cardiovascular regulation, and fitness in general. However, more mechanisms and more tissues are likely involved in the integrative response to habitual exercise. **Epigenetics** is the study of changes in gene expression that occur without changing the genetic code itself. For example, inherited factors clearly influence an individual's response to exercise.

However, additional environmental factors can alter those genes by epigenetic modifications, which are changes in how genes function that do not change the nucleotide sequence of the genes themselves. Environmental stimuli can alter the epigenome in a stable and inheritable fashion. Epigenetic modifications include DNA methylation, histone modification, and noncoding RNAs.²⁰ Although this area of research is relatively new, recent studies have demonstrated that epigenetic modifications contribute to altered gene expression in response to regular exercise; these findings have implications for improving our understanding of exercise-induced health benefits. The field of exercise epigenetics is still in its infancy but will increasingly provide new insights into human adaptations to exercise.

Bioinformatics

The techniques described in the previous sections generate an enormous amount of complex data. Sophisticated technologies, computer software, and statistical methods are therefore critical to analyzing the vast amount of genetic and molecular data generated from a single study, not to mention the integration of information from tens, hundreds, or thousands of experiments. **Bioinformatics** is essentially the management information system for molecular biology, serving as an intersection between molecular data and advanced mathematical and statistical approaches.¹⁸

Bioinformatics techniques allow us to address physiological questions that are otherwise unattainable using conventional methods. Using robotics, software for data processing and control, liquid handling devices, and sensitive detectors, high-throughput biology allows researchers to quickly conduct millions of chemical, genetic, or pharmacological tests by automating experiments on a large scale. By doing this, it becomes feasible to repeat experiments thousands of times. Using high-throughput methods, we can rapidly identify active compounds, antibodies, or genes that control or alter a particular physiological pathway.

Over the past decade, much has been learned from applying -omics approaches to the field of exercise physiology, and as this area of research continues to advance, bioinformatics will continue

to play a role. The development of software-based analyses that would consider the genetic profile of an individual and then predict his or her response to aerobic exercise training is one example of a potential application of bioinformatics and functional -omics in exercise physiology. As more and more research laboratories begin to incorporate -omics approaches in exercise physiology, the need for the bioinformatics tools to analyze and interpret the data will only increase.

One of the main goals of exercise physiology in the 21st century is to map function from **genotype** (the genetic makeup of an individual) to **phenotype** (observable characteristics of an individual resulting from the interaction of its genotype with the environment). In essence, exercise is a powerful stimulus that influences gene transcription across multiple tissues with implications for multiple phenotypes. It is tempting to speculate that, in the future, perhaps a person's genotype will be fed into an algorithm that can make predictions about exercise-related attributes, such as endurance, speed, strength, or adaptability. From there, an individualized and optimized training program could be developed.

However, it is important to remember that, while these reductionist methods and -omics approaches have provided important new information about the genes and pathways that underlie the physiological responses to exercise, a much more comprehensive understanding of the complex interaction among various genetic and epigenetic factors is required to fully optimize the use of exercise for disease prevention and treatment.

Exercise Physiology Beyond Earth's Boundaries

An important segment of exercise physiology concerns the response and adaptation of people to extremes of heat, cold, and altitude. Understanding and controlling the physiological stresses and adaptations that occur at these environmental limits have contributed directly to notable societal achievements such as construction of the Brooklyn Bridge, the Hoover Dam, pressurized aircraft, and underwater habitats for the commercial diving industry.

The next generation of environmental challenges will also require such physiological expertise. Commercial space vehicles now travel

routinely to low Earth orbit. NASA recently announced a set of new initiatives that will place humans in deep orbits near the moon in the late 2020s, followed by regular trips to Martian orbit in the 2030s. Indeed, we are on the verge of becoming an interplanetary civilization!

There are tremendous physiological and psychological challenges imposed on humans living in space and on planetary bodies for extended periods of time. The continuous pull of gravity contributes to the growth and adaptation of postural skeletal muscles, loads bones, which increases their size and density, and requires the cardiovascular system to maintain blood pressure and brain blood flow. In a microgravity environment (free fall around the Earth or constant-velocity conditions in deep space), the reduction in loading leads to dramatic losses in muscle mass and strength, osteoporosis, and exercise intolerance at rates that mimic those seen in spinal cord-injured patients.

Beginning in the 1980s, experiments done aboard a series of dedicated space shuttle flights investigated these problems in detail. The National Aeronautics and Space Administration (NASA) began flying the European Space Agency-developed Spacelab module, ushering in a new era of internationally sponsored scientific research into low Earth orbit. The Spacelab Life Sciences (SLS-1, SLS-2) missions (STS-40 and STS-58) emphasized the study of cardiorespiratory, vestibular, and musculoskeletal adaptations to microgravity, and the Life and Microgravity Sciences mission (STS-78) concentrated on neuromuscular adaptation. The 1998 Neurolab mission (STS-90), with an exclusive neuroscience theme, concluded flights of the Spacelab module. Dr. James A. Pawelczyk, a Penn State exercise physiologist and payload specialist on that flight, cotaught the first exercise physiology class from space!

With the end of the space shuttle program, work continues today aboard the International Space Station, which has provided a continuous human presence in space for nearly 20 years. The tools of modern molecular biology are helping to elucidate how loading, radiation, and stress interact to affect all physiological systems.

For the exercise physiologist, the question is what combination of resistance and aerobic exercise training can prevent or diminish the

changes that occur during space exploration. At this time, the answer is not complete. Furthermore, if physical conditioning is required before, during, and after space missions that could last up to 30 months, how should exercise prescriptions be individualized, evaluated, and updated? Without doubt, further research in exercise and environmental physiology will be essential to complete what is destined to be the largest exploration feat of the 21st century.

Research: The Foundation for Understanding

Exercise and sport scientists actively engage in research to better understand the mechanisms that regulate the body's physiological responses to acute bouts of exercise, as well as its adaptations to training and detraining. Most of this research is conducted at major research universities, medical centers, and specialized institutes using standardized research approaches and select tools of the exercise physiologist.

The Research Process

Science and research (the process by which science is developed) involve a process designed to pose and answer appropriate questions, develop testable hypotheses, test those hypotheses appropriately, generate usable data, interpret those data, and either accept or refute the original hypotheses. The research process is illustrated in [figure 0.11](#). Scientists are constantly challenged to make careful observations either from nature or the scientific literature, then ask focused questions that can be examined using a well-designed and well-controlled experimental process. The usual result of this overall process is submitting a research manuscript to an appropriate scientific journal, where it is peer reviewed, revised, and (hopefully) published. As other scientists read the research paper, they may in turn craft their own follow-up questions, and the process continues.

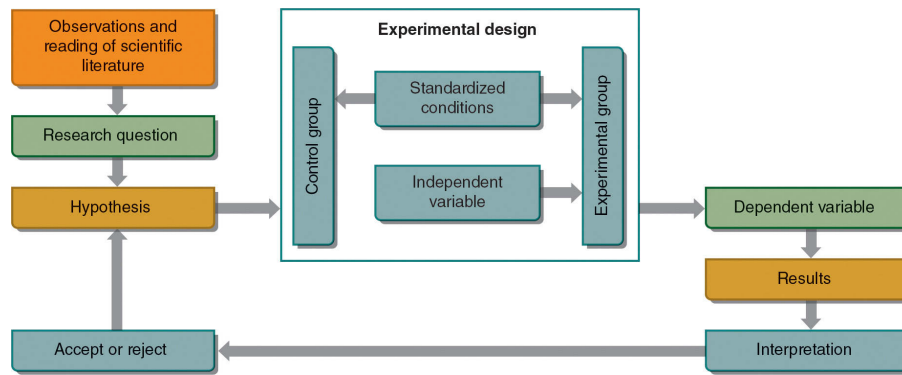


FIGURE 0.11 A simplified diagram of the typical process involved in scientific research.



Research Settings

Research can be conducted either in the laboratory or the field. Laboratory tests are usually more accurate because more specialized and sophisticated equipment can be used and conditions can be carefully controlled. As an example, the direct laboratory measurement of maximal oxygen uptake ($\dot{V}O_{2\max}$) is considered the most accurate estimate of cardiorespiratory endurance capacity. However, some field tests, such as the 1.5 mi (2.4 km) run, are also used to estimate $\dot{V}O_{2\max}$. These field tests, which measure the time it takes to run a set distance or the distance that can be covered in a fixed time, are not totally accurate, but they provide a reasonable estimate of $\dot{V}O_{2\max}$, are inexpensive to conduct, and allow many people to be tested in a short time. Field tests can be conducted in the workplace, on a running track or in a swimming pool, or during athletic competitions. To measure $\dot{V}O_{2\max}$ directly and accurately, one would need to go to a university or clinical laboratory.

Research Tools: Ergometers

When physiological responses to exercise are assessed in a laboratory setting, the participant's physical effort must be controlled to provide a measurable exercise intensity. This is generally accomplished through use of ergometers. An **ergometer** (*ergo* = work; *meter* = measure) is an exercise device that allows the intensity of exercise to be controlled (standardized) and measured.

Treadmills

Treadmills are the ergometers of choice for most researchers and clinicians, particularly in the United States. With these devices, a motor drives a large belt on which a subject can either walk or run; thus, these ergometers are often called motor-driven treadmills (see [figure 0.12](#)). Belt length and width must accommodate the individual's body size and stride length. For example, it is nearly impossible to test elite athletes on treadmills that are too short, or obese subjects on treadmills that are too narrow or not sturdy enough.

Treadmills offer a number of advantages. Walking is a natural activity for almost everyone, so individuals normally adjust to the skill required for walking on a treadmill within a few minutes. Also, most people can achieve their peak values for most physiological variables (heart rate, ventilation, oxygen uptake) on the treadmill, although some athletes (e.g., competitive cyclists) achieve higher values on ergometers that more closely match their mode of training or competition.



FIGURE 0.12 A motor-driven treadmill.

Treadmills do have some disadvantages. They are generally more expensive than simpler ergometers, like the cycle ergometers discussed next. They are also bulky, require electrical power, and are not very portable. Accurate measurement of blood pressure during treadmill exercise can be difficult because both the noise associated with normal treadmill operation and subject movement can make hearing through a stethoscope difficult.

Cycle Ergometers

For many years, the **cycle ergometer** was the primary testing device in use, and it is still used extensively in both research and clinical settings. Cycle ergometers can be designed to allow subjects to pedal either in the normal upright position (see [figure 0.13](#)) or in reclining or semireclining positions.

Cycle ergometers in a research setting generally use either mechanical friction or electrical resistance. With mechanical friction devices, a belt encompassing a flywheel is tightened or loosened to

adjust the resistance against which the cyclist pedals. The power output depends on the combination of the resistance and the pedaling rate—the faster one pedals, the greater the power output. To maintain the same power output throughout the test, one must maintain the same pedaling rate, so pedaling rate must be constantly monitored.



FIGURE 0.13 A cycle ergometer.

With electrically braked cycle ergometers, the resistance to pedaling is provided by an electrical conductor that moves through a magnetic or electromagnetic field. The strength of the magnetic field determines the resistance to pedaling. These ergometers can be controlled so that the resistance increases automatically as pedal rate decreases, and decreases as pedal rate increases, to provide a constant power output.

Similar to treadmills, cycle ergometers offer some advantages and disadvantages compared to other ergometers. Exercise intensity on a cycle ergometer does not depend on the subject's body weight.

This is important when one is investigating physiological responses to a standard rate of work (power output). As an example, if someone lost 5 kg (11 lb), data derived from treadmill testing could not be compared with data obtained before the weight loss because physiological responses to a set speed and grade on the treadmill vary with body weight. After the weight loss, the rate of work at the same speed and grade would be less than before. With the cycle ergometer, weight loss does not have as great an effect on physiological response to a standardized power output. Thus, walking or running is often referred to as weight-dependent exercise, while cycling is weight independent.

Cycle ergometers also have disadvantages. If the subject does not regularly engage in that form of exercise, the leg muscles will likely fatigue early in the exercise bout. This may prevent a subject from attaining a true maximal intensity. When exercise is limited in this way, responses are often referred to as peak exercise intensity rather than maximal exercise intensity. This limitation may be attributable to local leg fatigue, blood pooling in the legs (less blood returns to the heart), or the use of a smaller muscle mass during cycling than during treadmill exercise. Trained cyclists, however, tend to achieve their highest peak values on the cycle ergometer.

Other Ergometers

Other ergometers allow athletes who compete in specific sports or events to be tested in a manner that more closely approximates their training and competition. For example, an arm ergometer may be used to test athletes or nonathletes who use primarily their arms and shoulders in physical activity. Arm ergometry has also been used extensively to test and train athletes paralyzed below arm level. The rowing ergometer was devised to test competitive rowers.

Valuable research data have been obtained by instrumenting swimmers and monitoring them during swimming in a pool. However, the problems associated with turns and constant movement led to the use of two devices—tethered swimming and swimming flumes. In tethered swimming, the swimmer is attached to a harness connected to a rope, a series of pulleys, and counterbalancing weights and must swim against the pull of the apparatus to maintain

a constant position in the pool. A swimming flume allows swimmers to more closely simulate their natural swimming strokes. The swimming flume operates by pumps that circulate water past the swimmer, who attempts to maintain body position in the flume. The pump circulation can be increased or decreased to vary the speed at which the swimmer must swim. The swimming flume, which unfortunately is very expensive, has at least partially resolved the problems with tethered swimming and has created new opportunities to investigate the sport of swimming.

When one is choosing an ergometer, the concept of specificity is particularly important with highly trained athletes. The more specific the ergometer is to the actual pattern of movement used by the athlete in his or her sport, the more meaningful will be the test results.

In Review

- Treadmills generally produce higher peak values than other ergometers for almost all assessed physiological variable, such as heart rate, ventilation, and oxygen uptake.
- Cycle ergometers are the most appropriate devices for evaluating changes in submaximal physiological function before and after training in people whose weights have changed. Unlike treadmill exercise, cycle ergometer intensity is largely independent of body weight.

Research Designs

In exercise physiology research, there are two basic types of research design: cross-sectional and longitudinal. With a **cross-sectional research design**, a cross section of the population of interest (that is, a representative sample) is tested at one specific time, and the differences between subgroups from that sample are compared. With a **longitudinal research design**, the same research subjects are retested periodically after initial testing to measure changes over time in variables of interest.

The differences between these two approaches are best understood through an example. The objective of a research study is to determine whether a regular program of distance running increases the concentration of cardioprotective high-density

lipoprotein cholesterol (HDL-C) in the blood. High-density lipoprotein cholesterol is the desirable form of cholesterol; increased concentrations are associated with reduced risk for heart disease. Using the cross-sectional approach, one could, for example, test a large number of people who fall into the following categories:

- A group of subjects who do no training (the control group)
- A group of subjects who run 24 km (15 mi) per week
- A group of subjects who run 48 km (30 mi) per week
- A group of subjects who run 72 km (45 mi) per week
- A group of subjects who run 96 km (60 mi) per week

One would then compare the results from all the groups, basing one's conclusions on how much running was done. Using this approach, exercise scientists found that weekly running results in elevated HDL-C levels, suggesting a positive health benefit related to running distance. Furthermore, as illustrated in [figure 0.14](#), there was a **dose-response relation** between these variables—the higher the dose of exercise training, the higher the resulting concentration of HDL-C. It is important to remember, however, that with a cross-sectional design, these are different groups of runners, not the same runners at different training volumes.

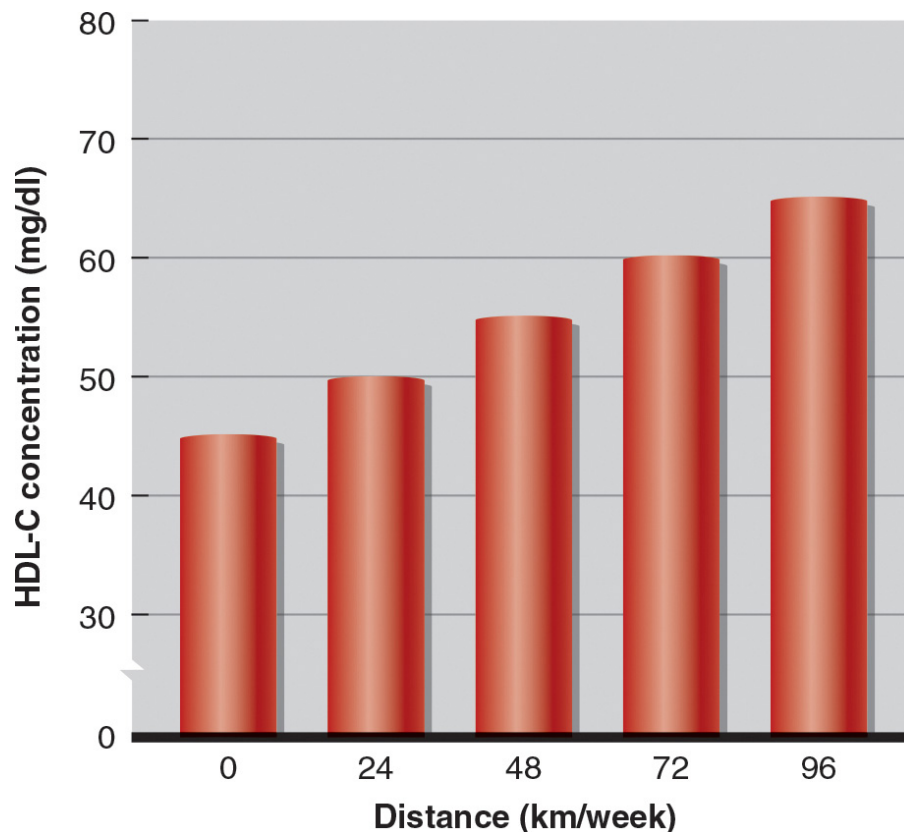


FIGURE 0.14 The relation between distance run per week and average high-density lipoprotein cholesterol (HDL-C) concentrations across five groups: nontraining control (0 km/week), 24 km/week, 48 km/week, 72 km/week, and 96 km/week. This illustrates a cross-sectional study design.



Using the longitudinal approach to test the same question, one could design a study in which untrained people would be recruited to participate in a 12-month distance-running program. One could, for example, recruit 40 people willing to begin running and then randomly assign 20 to a training group and the remaining 20 to a **control group**. Both groups would be followed for 12 months. Blood samples would be tested at the beginning of the study and then at 3-month intervals, concluding at 12 months when the program ended. With this design, both the running group and the control group would be followed over the entire period of the study, and changes in their

HDL-C levels could be determined across each period. Studies have been conducted using this longitudinal design to examine changes in HDL-C with training, but their results have not been as clear as the results of the cross-sectional studies. See [figure 0.15](#) as an example. Note that in this figure, in contrast to [figure 0.14](#), there is only a small increase in HDL-C in the subjects who are training. The control group stays relatively stable, with only minor fluctuations in their HDL-C from one 3-month period to the next.

A longitudinal research design is usually best suited to studying changes in variables over time. Too many factors that may taint results can influence cross-sectional designs. For example, genetic factors might interact so that those who perform well in long-distance running are also those who have high HDL-C levels. Also, different populations might follow different diets. In a longitudinal study, diet and other variables can be more easily controlled. However, longitudinal studies are time consuming and expensive to conduct, and are not always possible; cross-sectional studies do provide some insight into the questions at hand.

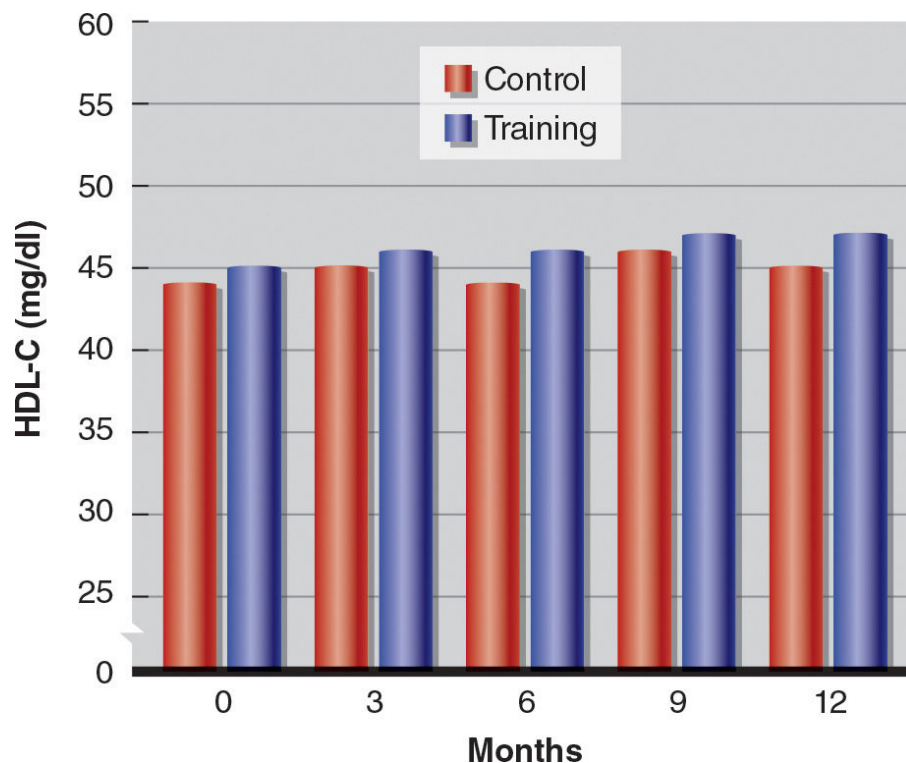


FIGURE 0.15 The relation between months of distance-running training and average high-density lipoprotein cholesterol (HDL-C) concentrations in an experimental group (20 subjects, distance training)

and a sedentary (20 subjects) control group. This illustrates a longitudinal study design.



Research Controls

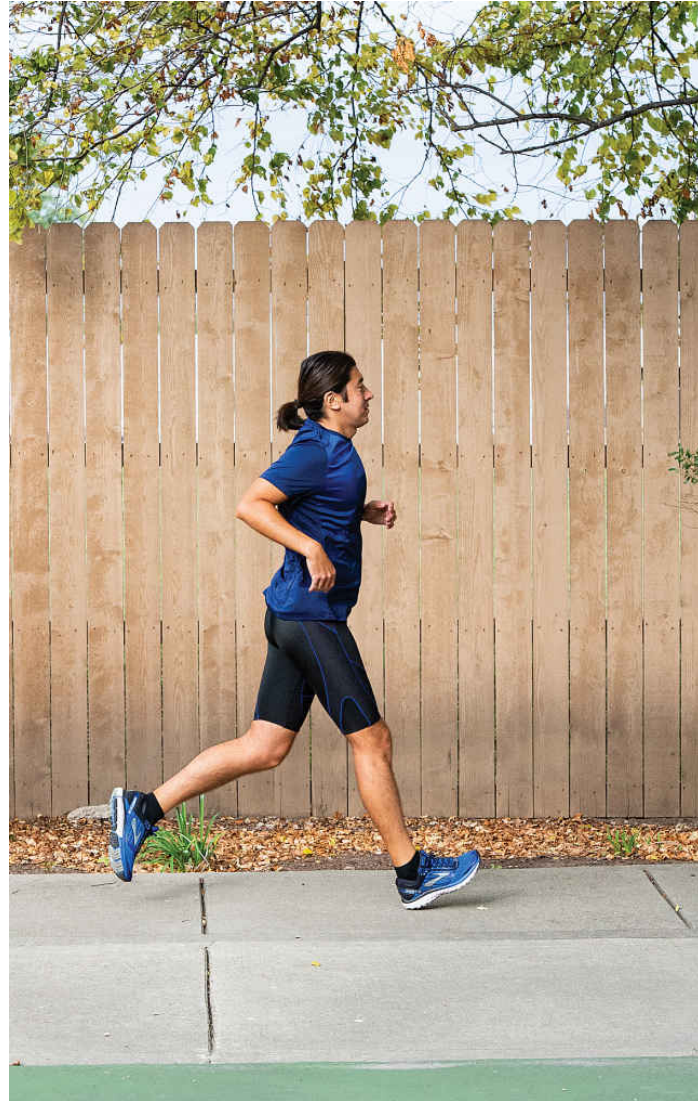
When we conduct research, it is important to be as careful as possible in designing the study and collecting the data. We see from [figure 0.15](#) that changes in a variable over time resulting from an intervention such as exercise can be very small. Yet, even small changes in a variable such as HDL-C can mean a substantial reduction in risk for heart disease. Recognizing this, scientists design studies aimed at providing results that are both accurate and reproducible. This requires that studies be carefully controlled.

Research controls are applied at various levels. Starting with the design of the research project, the scientist must determine how to control for variation in the subjects used in the study. The scientist must determine if it is important to control for the subjects' sex, age, or body size. To use age as an example, for certain variables, the response to an exercise training program might be different for a child or an aged person compared with a young or middle-aged adult. Is it important to control for the subject's smoking or dietary status? Considerable thought and discussion are needed to make sure that the subjects used in a study are appropriate for the specific research question being asked.

For almost all studies, it is critical to have a control group. In the longitudinal research design for the cholesterol study described earlier, the **control group** acts as a comparison group to make certain that any changes observed in the running group are attributable solely to the training program and not to any other factors, such as the time of the year or aging of the subjects during the course of the study. Experimental designs often employ a **placebo group**. Thus, in a study in which a subject might expect to have a benefit from the proposed intervention, such as the use of a specific food or drug, a scientist might decide to use three groups of

subjects: an intervention group that receives the actual food or drug, a placebo group that receives an inert substance that looks exactly like the actual food or drug, and a control group that receives nothing. (The last group often serves as a time control, accounting for nonexperimentally induced changes that may occur over the course of the study period.) If the intervention and placebo groups improve their performance to the same level and the control group does not improve performance, then the improvement is likely the result of the placebo effect, or the expectation that the substance will improve performance. If the intervention group improves performance, and the placebo and control groups do not, then we can conclude that the intervention does improve performance.

One other way of controlling for the placebo effect is to conduct a study that uses a **crossover design**. In this case, each group undergoes both treatment and control trials at different times. For example, one group is administered the intervention for the first half of the study (e.g., 6 months of a 12-month study) and serves as a control during the last half of the study. The second group serves as a control during the first half of the study and receives the intervention during the second half. In some cases, a placebo can be used in the control phase of the study. [Chapter 16](#), Ergogenic Aids in Sport, provides further discussion of placebo groups.



It is equally important to control data collection. The equipment must be calibrated so the researcher knows that the values generated by a given piece of equipment are accurate, and the procedures used in collecting data must be standardized. For example, when using a scale to measure the weight of subjects, researchers need to calibrate that scale by using a set of calibrated weights (e.g., 10 kg, 20 kg, 30 kg, and 40 kg) that have been measured on a precision scale. These weights are placed on the scale to be used in the study, individually and in combination, at least once a week to provide certainty that the scale is measuring the weights accurately. As another example, electronic analyzers used to measure respiratory gases need to be calibrated frequently with

gases of known concentration to ensure the accuracy of these analyses.

Finally, it is important to know that all test results are reproducible. In the example illustrated in [figure 0.15](#), the HDL-C of an individual is measured every 3 months. If that person is tested 5 days in a row before he or she starts the training program, one would expect the HDL-C results to be similar across all 5 days, provided that diet, exercise, sleep, and time of day for testing remained the same. In [figure 0.15](#), the values for the control group across 12 months varied from about 44 to 45 mg/dl, whereas the exercise group values increased from 45 to 47 mg/dl. Over five consecutive days, the measurements should not vary by more than 1 mg/dl for any one person if the researcher is going to pick up this small change over time. To control for reproducibility of results, scientists generally take several measurements, sometimes on different days, and then average the results before, during, and at the end of an intervention.

Confounding Factors in Exercise Research

Many factors can alter the body's acute response to a bout of exercise. For example, environmental conditions such as the temperature and humidity of the laboratory and the amount of light and noise in the test area can markedly affect physiological responses, both at rest and during exercise. Even the timing, volume, and content of the last meal and the quantity and quality of sleep the night before must be carefully controlled in research studies.

To illustrate this, [table 0.1](#) shows how varying environmental and behavioral factors can alter heart rate at rest and during running on a treadmill at 14 km/h (9 mph). The subject's heart rate response during exercise differed by 25 beats/min when the air temperature was increased from 21 °C (70 °F) to 35 °C (95 °F). Most physiological variables that are normally measured during exercise are similarly influenced by environmental fluctuations. Whether one is comparing a person's exercise results from one day to another or comparing the responses of two different subjects, all of these factors must be controlled as carefully as possible.

Physiological responses, both at rest and during exercise, also vary throughout the day. The term **diurnal variation** refers to fluctuations that occur during a 24 h day. Because such variables as body temperature and heart rate vary naturally during a 24 h period, testing the same person in the morning on one day and in the afternoon on the next will produce different results. Test times must be standardized to control for this diurnal effect.

TABLE 0.1 Heart Rate Responses to Running Differ with Variations in Environmental and Behavioral Conditions

Environmental and behavioral factors	Heart rate (beats/min)	
	Rest	Exercise
Temperature (at 50% humidity)		
21 °C (70 °F)	60	165
35 °C (95 °F)	70	190
Humidity (at 21 °C)		
50%	60	165
90%	65	175
Noise level (at 21 °C, 50% humidity)		
Low	60	165
High	70	165
Food intake (at 21 °C, 50% humidity)		
Small meal 3 h before exercising	60	165
Large meal 30 min before exercising	70	175
Sleep (at 21 °C, 50% humidity)		
8 h or more	60	165
6 h or less	65	175

At least one other physiological cycle must also be considered. The normal 28-day menstrual cycle often involves considerable variations in

- body weight,
- total body water and blood volume,
- body temperature,
- metabolic rate, and
- heart rate and stroke volume (the amount of blood leaving the heart with each contraction).

Exercise scientists must control for menstrual cycle phase or the use of oral contraceptives (which similarly alter hormonal status), or both, when testing women. When older women are being tested, testing strategies must take into account menopause and hormone replacement therapies.

In summary, the conditions under which research participants are monitored, at rest and during exercise, must be carefully controlled. Environmental factors, such as temperature, humidity, altitude, and noise, can affect the magnitude of response of all basic physiological systems, as can behavioral factors such as eating patterns and sleep. Likewise, physiological measurements must be well controlled for diurnal and menstrual cycle variations.

Units and Scientific Notation

A set of international standards for units and abbreviations (SI, Le Système International d'Unités) serves as the preferred units of measurement in exercise and sport physiology. In this text, alternate units in common use (such as weight in pounds) are often provided as well. Many of these units are provided on the inside front cover of this text, and conversions for units in common use are found on the inside back cover.

In common writing and even in mathematics, the ratio between two numbers is typically written using a slash (/). For example, in dry air at 20 °C, the speed of sound is 343 m/s. That notation works well for simple fractions or ratios, and we have maintained it in this text. However, the notation gets confusing for relations of several—that is, more than two—variables. Take, for instance, one of the cornerstone measurements in exercise physiology, an individual's maximal oxygen uptake or maximal aerobic capacity, abbreviated $\dot{V}O_{2\max}$. This important physiological measurement is the maximal volume of oxygen that an individual can use during exhaustive aerobic exercise, and can be measured in liters per minute, or L/min. However, because a large person can use more oxygen yet not be more aerobically fit, we often standardize this value to body weight in kilograms, that is, milliliters per kilogram per minute. Now the notation becomes a bit more complex and potentially more confusing. We could write the units as ml/kg/min, but what is being divided by what in this notation? Recall that L/min can also be written as $L \cdot \text{min}^{-1}$, just as the fraction $1/4 = 1 \cdot 4^{-1}$. To avoid errors and ambiguity, in exercise physiology, we use the exponent notation any time more than two variables are involved. Therefore, milliliters per

kilogram per minute is written as $\text{ml} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$ rather than ml/kg/min .

Reading and Interpreting Tables and Graphs

This book contains references to specific research studies that have had a major impact on our understanding of exercise and sport physiology. Once scientists complete a research project, they submit the results of their research to one of the many research journals in sport and exercise physiology.

As in other areas of science, most of the quantitative research in exercise physiology is presented in the form of tables and graphs. Tables and graphs provide an efficient way for researchers to communicate the results of their studies to other scientists. For the student in exercise and sport physiology, a working knowledge of how to read and interpret tables and graphs is critical.

Tables are usually used to convey a large number of data points or complex data that are affected by several factors. Take [table 0.1](#) as an example. It is important to first look at the title of the table, which identifies what information is being presented. In this case, the table is designed to illustrate how various conditions affect heart rate, at rest and during exercise. The left-hand column, along with the horizontal subheadings (like “Humidity (at 21 °C)”), specify the conditions under which the heart rate was measured. Columns 2 and 3 provide the mean heart rate values that correspond to each condition, with the middle column giving the resting value and the right-most column the exercise value. In every good table and graph, the units for each variable are clearly presented; in this table, heart rate is expressed in beats/min, or beats per minute. Pay careful attention to the units of measure used when interpreting a table or graph. From this table—a relatively simple one—we see that both resting and exercise heart rate were increased by increased ambient temperature and humidity, while noise level affected only resting heart rate. Similarly, consuming a large meal or getting less than 6 h of sleep also raises heart rate. These data could not easily have been shown in graphical form.

Graphs can provide a better view of trends in data, response patterns, and comparisons of data collected from two or more groups

of subjects. For some students, graphs can be more difficult to read and interpret, but graphs are, and will remain, a critical tool in the understanding of exercise physiology. First, every graph has a horizontal axis, or x-axis, for the **independent variable** and a vertical axis, or y-axis, (or sometimes two) for the **dependent variable** or variables. Independent variables are those factors that are manipulated or controlled by the researcher, while dependent variables are those that change with—that is, depend on—the independent variables.

In [figure 0.16](#), time of day is the independent variable and is therefore placed along the x-axis of the graph, while heart rate is the dependent variable (since heart rate *depends on* the time of day) and is therefore plotted on the y-axis. The units of measure for each variable are clearly displayed on the graph. [Figure 0.16](#) is in the form of a line graph. Line graphs are useful in illustrating patterns or trends in data but should be used only to compare two variables that change in a continuous manner (for example, across time) and only if both the dependent and independent variables are numbers.

In a line graph, if the dependent variable goes up or down at a constant rate with the independent variable, the result will be a straight line. However, in physiology, the response pattern between variables is often not a straight line but a curve of one shape or another. In such cases, pay close attention to the slope of various parts of the curve as it changes across the graph. For instance, [figure 0.17](#) shows the concentration of lactate in the blood as subjects walk-run on a treadmill at various increasing speeds. At low treadmill speeds of 4 to 8 km/h, lactate increases very little. However, at about 8.5 km/h, a threshold is reached beyond which lactate increases more dramatically. In many physiological responses, both the threshold (onset of response) and the slope of the response beyond that threshold are important.

Data can also be plotted in the format of a bar graph. Bar graphs are commonly used when only the dependent variable is a number and the independent variable is a category. Bar graphs often show treatment effects, as in [figure 0.14](#), which was previously discussed. [Figure 0.14](#) shows the effect of distance run per week (a category) on HDL-C (a numerical response) in the bar graph format.

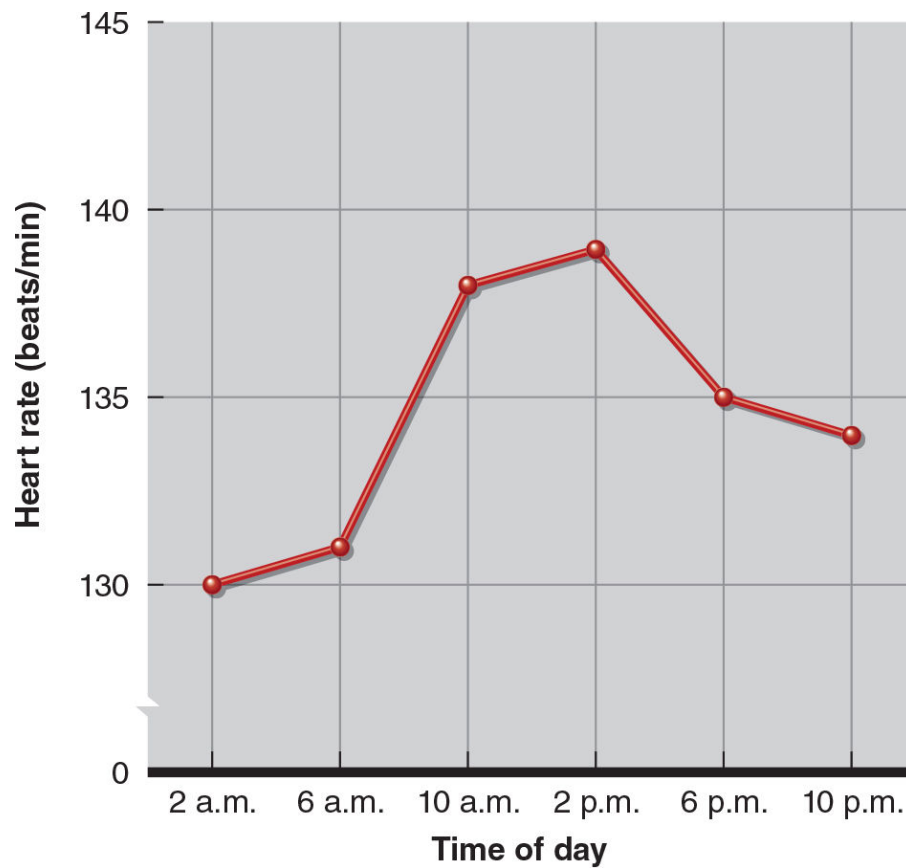


FIGURE 0.16 This line graph depicts the relation between the time of day (on the x-axis, independent variable) and heart rate during low-intensity exercise (on the y-axis, dependent variable) that was measured at that time of day with no change in the exercise intensity.



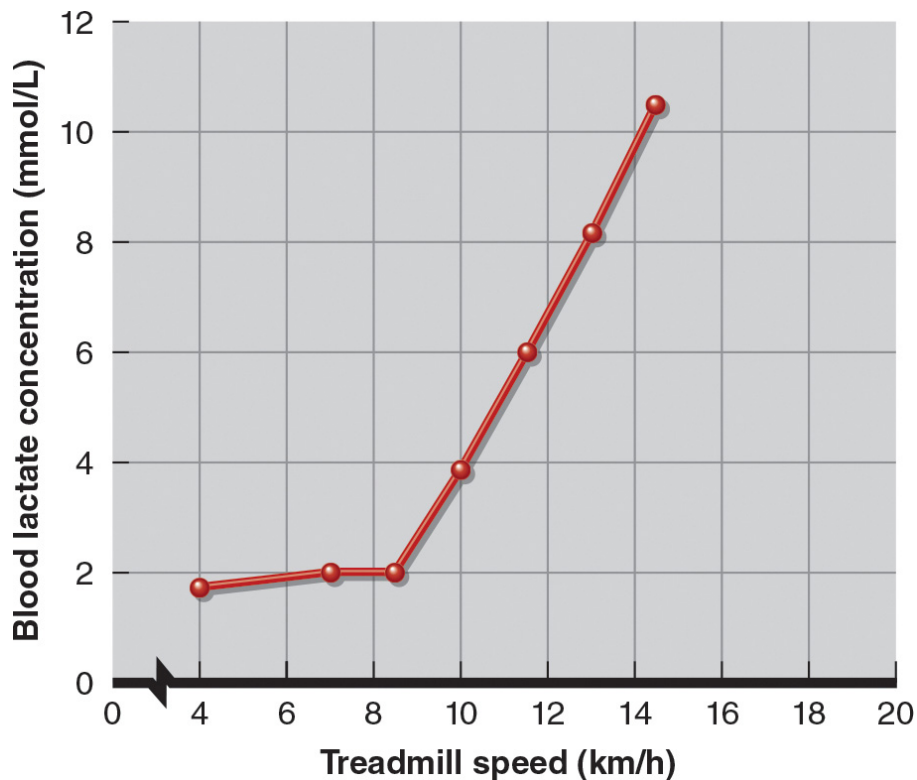


FIGURE 0.17 A line graph showing the nonlinear nature of many physiological responses. This graph shows that above a threshold (onset of response) of about 8.5 km/h, the slope of the blood lactate response increases sharply.



In Review

- Exercise physiologists make use of both cross-sectional (finding differences between groups at one point in time) and longitudinal (retesting the same subjects at different points in time) research designs.
- For all sound research studies, it is critical to have a control group as well as an experimental group. The control group often involves a placebo treatment rather than no treatment at all.
- Exercise physiologists use the SI units of measurement and abbreviations.

IN CLOSING

In this introduction, we highlighted the historical roots and scientific underpinnings of exercise and sport physiology. We learned that the current state of knowledge in these fields builds on the past and is a bridge to the future—many questions remain unanswered. New and exciting approaches and techniques are being developed continually. While reductionist (e.g., genomics) approaches are growing in popularity, being able to integrate those findings into a systems and whole-body perspective will never go out of style. Exercise and sport physiology is an important part of integrative and translational physiology. We briefly defined the acute responses to exercise bouts and chronic adaptations to long-term training. We concluded with an overview of the principles used in sport and exercise physiology research as well as an introduction to interpreting graphs, some important terminology, and SI units and their notation.

In [part I](#), we begin examining physical activity the way exercise physiologists do as we explore the essentials of movement. In the next chapter, we examine the structure and function of skeletal muscle, how it produces movement, and how it responds during exercise.

KEY TERMS

acute exercise
bioinformatics
chronic adaptation
control group
crossover design
cross-sectional research design
cycle ergometer
dependent variable
diurnal variation
dose–response relation
environmental physiology
epigenetics
ergometer
exercise physiology
genomics
genotype
homeostasis
independent variable
integrative physiology
longitudinal research design

phenotype
physiology
placebo group
sport physiology
training effect
translational physiology
treadmill

STUDY QUESTIONS

1. What is exercise physiology? How does sport physiology differ?
2. Provide an example of “studying acute responses to a single bout of exercise.”
3. Describe what is meant by “studying chronic adaptations to exercise training.”
4. Describe the evolution of exercise physiology from the early studies of anatomy. Who were some of the key figures in the development of this field?
5. Describe the founding and the key areas of research emphasized by the Harvard Fatigue Laboratory. Who was the first research director of this laboratory?
6. Name the three Scandinavian physiologists who conducted research in the Harvard Fatigue Laboratory.
7. What is an ergometer? Name the two most commonly used ergometers and explain their advantages and disadvantages.
8. What factors must researchers consider when designing a research study to ensure that they get accurate and reproducible results?
9. What is translational physiology?
10. Define the following terms and discuss their relevance to exercise physiology: genomics, epigenetics, bioinformatics, genotype, and phenotype.
11. List several environmental conditions that could affect one’s response to an acute bout of exercise.
12. What are the advantages and disadvantages of a cross-sectional versus a longitudinal study design?
13. When should data be depicted as a bar graph as opposed to a line graph? What purpose does a line graph serve?

STUDY GUIDE ACTIVITIES

In addition to the activities listed in the chapter opening outline, two other activities are available in the web study guide, located at

www.HumanKinetics.com/PhysiologyOfSportAndExercise

The **KEY TERMS** activity reviews important terms, and the end-of-chapter **QUIZ** tests your understanding of the material covered in the chapter.